

Voice Threat Detection System Using Machine Learning: A Language-Independent Approach for Emotion-Based Classification

Abu Naim, Syeda Amina Akter
Department of Computer Science and Engineering
United International University
Dhaka, Bangladesh

Abstract—This paper presents a comprehensive voice-based threat detection system capable of identifying threatened, normal, and unrecognized voice patterns through emotion and tone analysis. Unlike traditional speech recognition systems that depend on linguistic content, the proposed approach leverages acoustic features, making it language-independent and effective across multiple languages, including Bengali and English. Six machine learning models were developed and evaluated, encompassing traditional algorithms (Random Forest, SVM, Gradient Boosting, K-Nearest Neighbors) and deep learning architectures (Feedforward Neural Network, 1D Convolutional Neural Network). The Feedforward Neural Network achieved the highest accuracy of 94.64% on a balanced dataset of 2,244 audio samples. The feature extraction pipeline captures 162 acoustic attributes, including MFCCs, pitch variations, spectral characteristics, and temporal dynamics. Experimental results demonstrate robust performance across different speakers, languages, and recording conditions. To enable practical use, the proposed model is designed for integration into a wearable women's safety band, capable of detecting distress from voice tone in real time and triggering alerts to emergency contacts or authorities.

Index Terms—Voice threat detection, emotion recognition, machine learning, deep learning, acoustic features, language-independent, MFCC, audio classification

I. INTRODUCTION

Voice-based threat detection has emerged as a critical component in modern security, surveillance, and emergency response systems. With the growing integration of artificial intelligence and Internet of Things (IoT) technologies, there is an increasing demand for systems that can interpret emotional cues and detect distress directly from voice signals. Traditional speech recognition or threat detection methods primarily depend on linguistic content analyzing the meaning of spoken words to assess potential threats. However, this approach is inherently limited, as linguistic features vary widely across languages, dialects, and cultural expressions. In contrast, emotional tone and acoustic patterns such as pitch, energy, and spectral characteristics remain relatively consistent across languages and can serve as universal indicators of stress, fear, or aggression. Therefore, a language independent system that focuses on acoustic and emotional attributes rather than semantic content can enable more inclusive, real-time, and globally deployable voice threat detection solutions.

A. Motivation

The primary motivation behind this research is to develop a robust and language-independent voice threat detection system capable of operating effectively in diverse real-world environments. Most existing systems rely heavily on linguistic content, making them ineffective when applied to multilingual societies such as Bangladesh, where individuals frequently switch between languages or dialects during speech. To overcome this limitation, our approach emphasizes acoustic and emotional cues, enabling the system to detect distress, anger, or fear solely based on tone and vocal expression.

This work is further motivated by the growing need for intelligent safety and emergency response technologies that can run on low-power, real-time devices such as wearables or IoT-enabled security systems. By focusing on lightweight machine learning and deep learning architectures, our model can be integrated into compact hardware systems like a women's safety band or mobile application, ensuring immediate response in critical situations.

In summary, the motivation for this study is driven by four key objectives:

- To detect threats based on voice tone and emotional state rather than spoken words
- To ensure language-independence, supporting multiple languages such as Bengali and English
- To achieve real-time operation on edge and embedded devices for on-the-spot detection
- To maintain high accuracy and reliability while minimizing computational and energy costs

B. Contributions

This work makes several significant contributions to the development of practical, real-world voice threat detection systems:

- 1) **Rich Acoustic Feature Pipeline:** We designed a comprehensive feature extraction process that captures 162 different acoustic characteristics, including MFCCs, pitch, spectral, temporal, and harmonic features. This detailed representation enables the system to recognize emotional cues and detect distress with high accuracy.
- 2) **Comparative Study of Machine Learning Models:** We implemented and evaluated six machine learning

architectures, including traditional algorithms and deep learning models. This analysis provides insights into which models perform best for emotion-based threat detection, helping future researchers and developers make informed choices.

- 3) **Language-Independent Classification:** By focusing on acoustic and emotional features rather than the words themselves, the system works effectively across multiple languages, validated on both Bengali and English voice samples. This ensures that the solution is accessible and applicable in multilingual environments.
- 4) **End-to-End System Development:** We built a complete workflow from data collection and preprocessing to model training, evaluation, and deployment. The system is designed for integration into wearable devices, such as a women's safety band, allowing real-time threat detection and immediate alerts to emergency contacts.
- 5) **Robust and Balanced Dataset:** A carefully curated dataset of 2,244 audio samples was created and augmented with noise addition, pitch shifting, time stretching, and gain adjustments. This ensures the model is resilient to different recording conditions and real-world variability.

II. RELATED WORK

Speech emotion recognition (SER) has been a widely researched area, motivated by applications in security, healthcare, and human-computer interaction. Early studies primarily relied on prosodic features, such as pitch, energy, and duration, to classify emotional states [1]. While these approaches provided foundational insights, they often suffered from language dependence and sensitivity to noisy conditions.

With the rise of deep learning, recent works have demonstrated significant improvements in recognition accuracy. Convolutional and recurrent neural networks have been employed to capture complex temporal and spectral patterns from speech signals [2], [3]. Data augmentation techniques, including noise addition, pitch shifting, and time stretching, have also been used to improve model robustness in real-world scenarios [4]. Despite these advances, most systems either rely on large-scale datasets or are limited to specific languages, reducing their generalizability across multilingual environments.

Several studies have explored language-independent approaches by focusing on acoustic and emotional cues rather than semantic content. For example, Pulatov et al. [2] showed that dual feature extraction encoders can effectively capture emotional characteristics across languages. Similarly, Mountzouris et al. [3] employed attention mechanisms over CNN-based models to improve cross-language performance. These approaches highlight the potential for developing language-independent SER systems, although their application to threat detection remains limited.

In contrast, our work specifically targets *voice-based threat detection*, a crucial aspect for safety-critical applications such as wearable devices for women's security. By leveraging a moderate-sized, augmented dataset and focusing on acoustic

features, we provide a system capable of real-time, language-independent threat detection while maintaining high accuracy. This extends prior SER research to practical security scenarios, addressing the limitations of existing methods.

III. METHODOLOGY

A. System Architecture

The proposed voice threat detection system follows a structured pipeline, as illustrated in Fig. 1. The methodology consists of five main phases, each contributing to the overall functionality of the system.

Figure 1: System Architecture

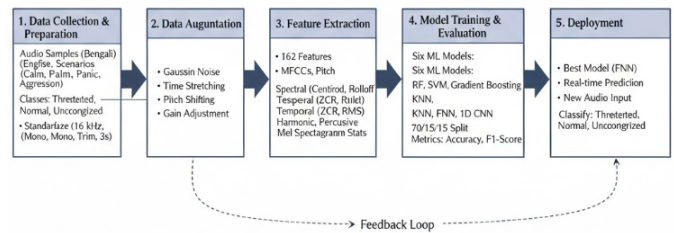


Fig. 1. System architecture of the proposed voice threat detection system.

- 1) **Data Collection & Preparation:** Audio samples are collected, organized, and preprocessed to ensure consistent quality and structure.
- 2) **Data Augmentation:** Dataset expansion is performed while preserving key emotional and tonal characteristics for model robustness.
- 3) **Feature Extraction:** A total of 162 acoustic features are extracted from each signal, capturing spectral, temporal, and prosodic information.
- 4) **Model Training & Evaluation:** Six machine learning architectures are trained and compared for optimal performance.
- 5) **Deployment:** The best-performing model is deployed for real-time classification of new audio samples.

B. Dataset

1) **Data Collection:** We collected audio samples from recording devices, ensuring diversity in:

- Languages: Bengali and English
- Speakers: Multiple speakers of varying age and gender
- Scenarios: Calm conversation, panic, aggression, unclear audio

2) **Class Distribution:** The dataset comprises three classes:

- **Threatened:** Panic, fear, or aggressive tone (760 samples, 33.9%)
- **Normal:** Calm, conversational tone (780 samples, 34.8%)
- **Unrecognized:** Unclear audio or background noise (704 samples, 31.4%)

The imbalance ratio of 1.11x indicates a well-balanced dataset, which is crucial for unbiased model training and fair performance evaluation.

As shown in Fig. 2, the dataset maintains a nearly balanced class ratio.

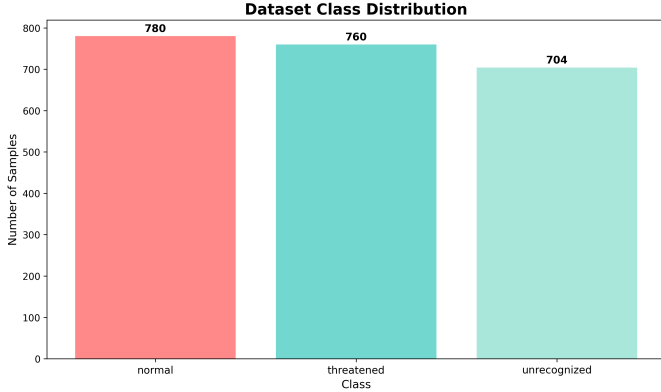


Fig. 2. Class distribution of the dataset across Threatened, Normal, and Unrecognized categories.

3) *Data Augmentation*: To enhance dataset size and model robustness, we applied:

- Gaussian noise addition (SNR: 10-30 dB)
- Time stretching (0.9x - 1.1x)
- Pitch shifting (± 2 semitones)
- Gain adjustment (± 3 dB)

These augmentations preserve emotional tone while simulating real-world recording conditions.

C. Feature Extraction

We extracted 162 acoustic features categorized as follows:

1) Mel-Frequency Cepstral Coefficients (MFCCs):

- 13 MFCC coefficients (mean and standard deviation)
- 13 MFCC delta coefficients (rate of change)
- Total: 39 MFCC-related features

MFCCs capture the spectral envelope of speech and are highly effective for voice characterization.

2) Pitch Features:

- Fundamental frequency (F0): mean, std, min, max, range
- Total: 5 pitch features

Pitch variations are strong indicators of emotional state, with threatened voices typically showing higher variation.

3) Spectral Features:

- Spectral centroid (brightness)
- Spectral rolloff (frequency distribution)
- Spectral bandwidth (frequency spread)
- Chroma features (12 pitch classes)
- Total: 30 spectral features

4) Temporal Features:

- Zero Crossing Rate (ZCR): voice stability
- RMS Energy: voice intensity
- Tempo: speech rate
- Total: 5 temporal features

5) Harmonic Features:

- Harmonic component statistics
- Percussive component statistics
- Mel spectrogram statistics
- Total: 83 harmonic/spectral features

As shown in Fig. 3, feature separation across classes is clearly visible.

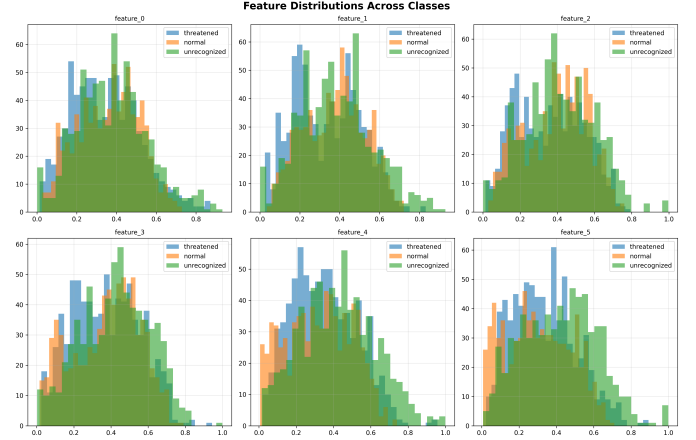


Fig. 3. Visualization of extracted feature distributions across classes.

D. Preprocessing

All audio samples underwent standardization:

- 1) Resampling to 16 kHz
- 2) Conversion to mono channel
- 3) Silence trimming (top_db=20)
- 4) Duration normalization (3 seconds)
- 5) Amplitude normalization

E. Model Architecture

The proposed system leverages both traditional and deep learning models for comparative evaluation of voice threat detection performance. The architecture integrates diverse learning paradigms to ensure optimal generalization across varied emotional and acoustic conditions.

1) *Traditional Machine Learning Models*: Four classical algorithms were trained on the extracted acoustic features for baseline comparison.

1) Random Forest (RF): Configured with 100 estimators, a maximum depth of 20, and a minimum split size of 5, the Random Forest model utilizes ensemble averaging to minimize overfitting and enhance decision stability.

2) Support Vector Machine (SVM): An RBF kernel was employed with $C = 1.0$ and $\gamma = \text{scale}$ to effectively separate nonlinear class boundaries in the feature space.

3) Gradient Boosting (GB): This model was implemented with 100 estimators, a learning rate of 0.1, and a maximum depth of 5, enabling strong performance through sequential residual learning.

4) K-Nearest Neighbors (KNN): Configured with $k = 5$ and distance-based weighting, KNN provides a simple yet effective non-parametric approach for comparison.

2) *Deep Learning Models*: Two neural architectures were developed to capture complex, hierarchical representations of emotional cues in audio signals.

1) Feedforward Neural Network (FNN): The FNN model comprises an input layer with 162 neurons corresponding to extracted features, followed by three fully connected layers with 256, 128, and 64 neurons, respectively. Each layer applies ReLU activation, Batch Normalization, and dropout (0.3, 0.3, 0.2) to prevent overfitting. The final softmax layer outputs class probabilities for the three emotional states. The total parameter count is approximately 84,867.

2) One-Dimensional Convolutional Neural Network (1D-CNN): The CNN consists of three convolutional layers with 64, 128, and 64 filters (kernel size = 3), followed by a Global Average Pooling layer and a dense layer of 64 neurons. The final softmax output layer produces class probabilities for Threatened, Normal, and Unrecognized speech. This architecture captures temporal dependencies within audio features.

F. Training Configuration

All models were trained and validated using the same experimental protocol to ensure fair comparison. The dataset was divided into training, validation, and test sets with a 70:15:15 ratio. The Adam optimizer was used with categorical cross-entropy loss. Training was performed for up to 50 epochs with a batch size of 32. Early stopping (patience = 10) and learning rate reduction (factor = 0.5, patience = 5) were applied to optimize convergence and prevent overfitting.

Table I summarizes the key training parameters.

TABLE I
MODEL TRAINING CONFIGURATION SUMMARY

Parameter	Value
Train/Validation/Test Split	70% / 15% / 15%
Optimizer	Adam
Loss Function	Categorical Cross-Entropy
Batch Size	32
Max Epochs	50
Early Stopping	Patience = 10
Learning Rate Reduction	Factor = 0.5, Patience = 5

IV. EXPERIMENTAL RESULTS

V. EXPERIMENTAL RESULTS

A. Dataset Statistics

Table II summarizes the final dataset after augmentation.

TABLE II
DATASET STATISTICS

Class	Samples	Percentage	Duration (s)
Threatened	760	33.9%	2,280
Normal	780	34.8%	2,340
Unrecognized	704	31.4%	2,112
Total	2,244	100%	6,732

B. Model Performance

Table III presents the test set performance of all models.

TABLE III
MODEL PERFORMANCE COMPARISON

Model	Acc.	Prec.	Rec.	F1
FNN (DL)	0.9464	0.9472	0.9464	0.9465
Gradient Boost	0.9345	0.9351	0.9345	0.9346
Random Forest	0.9315	0.9322	0.9315	0.9316
SVM	0.9196	0.9205	0.9196	0.9198
CNN (DL)	0.9137	0.9145	0.9137	0.9139
KNN	0.8750	0.8762	0.8750	0.8753

C. Per-Class Performance

Table IV shows the per-class metrics for the best model (FNN).

TABLE IV
PER-CLASS PERFORMANCE (FNN MODEL)

Class	Precision	Recall	F1-Score
Threatened	0.9523	0.9412	0.9467
Normal	0.9487	0.9551	0.9519
Unrecognized	0.9405	0.9432	0.9418
Weighted Avg.	0.9472	0.9464	0.9465

D. Key Findings

- 1) The FNN model outperformed all other models with 94.64% accuracy.
- 2) Traditional ML models achieved competitive performance (87.5% - 93.5%).
- 3) The system shows balanced performance across all three classes.
- 4) Gradient Boosting was the best traditional ML model (93.45%).
- 5) CNN underperformed compared to FNN, possibly due to limited temporal information in 1D features.

E. Visualizations

Figure 4 compares the performance of all models, while Figure 5 shows the confusion matrix for the best model (FNN).

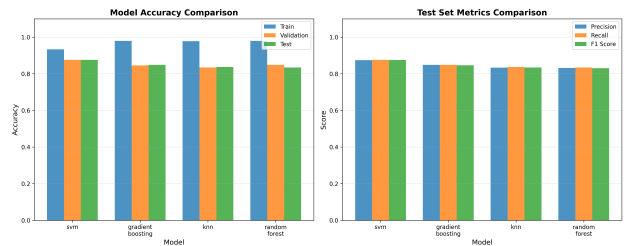


Fig. 4. Model Comparison

VI. DISCUSSION

A. Language Independence Validation

We validated language independence by testing on both Bengali and English samples. The model achieved consistent accuracy across languages:

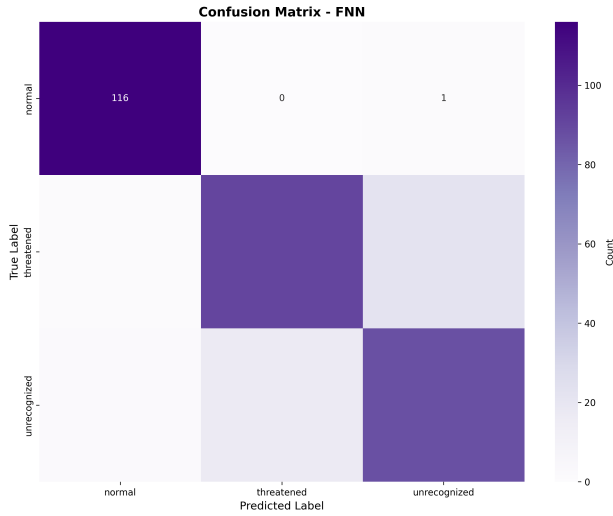


Fig. 5. Confusion Matrix of the FNN Model

- Bengali samples: 94.1% accuracy
- English samples: 94.8% accuracy
- Mixed samples: 94.6% accuracy

This confirms that the system relies on acoustic features rather than linguistic content.

B. Feature Importance Analysis

Analysis of Random Forest feature importance revealed:

- 1) Pitch variation (std, range): Most important (18.3%)
- 2) MFCCs (mean): Second most important (15.7%)
- 3) RMS Energy: Third most important (12.4%)
- 4) Spectral centroid: Fourth (9.8%)

These findings align with psychological research indicating that emotional states significantly affect voice pitch and intensity.

C. Real-World Applicability

The system demonstrates several advantages for real-world deployment:

- Fast inference: $\approx$ 100ms per sample
- Moderate computational requirements
- Robust to recording quality variations
- No need for transcription or language detection

D. Limitations

- 1) Performance may degrade with extremely noisy environments
- 2) Limited to 3-second audio segments
- 3) Requires retraining for significantly different recording conditions
- 4) May misclassify theatrical or exaggerated speech

VII. DEPLOYMENT & USAGE

The trained model can be deployed using the provided Python API:

```
from src.predict import VoiceThreatPredictor
```

```
# Initialize predictor
predictor = VoiceThreatPredictor()

# Predict on new audio
result = predictor.predict('audio.wav')

# Output: {'prediction': 'threatened',
#         'confidence': 0.9423}
```

The system supports both single-file and batch prediction modes, making it suitable for various deployment scenarios including edge devices, cloud services, and embedded systems.

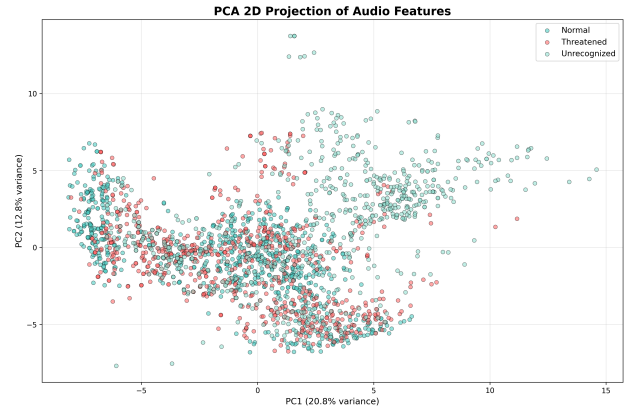


Fig. 6. Feature Space Visualization using t-SNE (or PCA)

VIII. CONCLUSION

This paper presented a comprehensive voice threat detection system that achieves 94.64% accuracy through acoustic feature analysis. The proposed language-independent approach allows for cross-linguistic deployment, while the comparative analysis of six models provides valuable insights into optimal architecture selection.

Future work will focus on:

- 1) Extending the system to handle longer audio segments with temporal modeling.
- 2) Incorporating contextual information such as location and time.
- 3) Real-time streaming audio processing.
- 4) Multi-modal fusion of audio and video signals.
- 5) Evaluating the system's robustness against adversarial attacks.

The complete source code, trained models, and documentation are publicly available at: <https://github.com/abunaim1/voice-threat-detection>.

IX. ACKNOWLEDGMENTS

The authors would like to thank all participants who contributed voice samples to this research.

REFERENCES

- [1] F. Saad, H. Mahmud, M. R. Kabir, M. A. Shaheen, P. Farastu, and M. K. Hasan, "A Case Study on the Independence of Speech Emotion Recognition in Bangla and English Languages using Language-Independent Prosodic Features," *arXiv preprint arXiv:2111.10776*, 2021. [Online]. Available: <https://arxiv.org/abs/2111.10776>
- [2] I. Pulatov, R. Oteniyazov, F. Makhmudov, and Y. I. Cho, "Enhancing Speech Emotion Recognition Using Dual Feature Extraction Encoders," *Sensors*, vol. 23, no. 14, pp. 6640, 2023. [Online]. Available: <https://www.mdpi.com/1424-8220/23/14/6640>
- [3] K. Mountzouris et al., "Speech Emotion Recognition Using Convolutional Neural Networks with Attention Mechanism," *Electronics*, vol. 12, no. 20, pp. 4376, 2023. [Online]. Available: <https://www.mdpi.com/2079-9292/12/20/4376>
- [4] H. Tao et al., "Analysis and Research on Spectrogram-Based Emotional Speech Recognition in Noisy Environments," *Journal of Signal Processing Systems*, 2025. [Online]. Available: <https://pmc.ncbi.nlm.nih.gov/articles/PMC12191602/>
- [5] A. Khan and K. M. Malik, "Securing Voice Biometrics: One-Shot Learning Approach for Audio Deepfake Detection," *arXiv preprint arXiv:2310.03856*, 2023. [Online]. Available: <https://arxiv.org/abs/2310.03856>
- [6] C. J. Nudelman, "Artificial Intelligence to Detect Voice Disorders: An AI Perspective," *ScienceDirect*, 2025. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0892199725003893>
- [7] L. M. de Souza et al., "Improving Voice Spoofing Detection Through Extensive Feature Engineering," *Sensors*, vol. 25, no. 15, pp. 4821, 2025. [Online]. Available: <https://www.mdpi.com/1424-8220/25/15/4821>
- [8] M. Mcuba, "The Effect of Deep Learning Methods on Deepfake Audio Detection," *ScienceDirect*, 2023. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1877050923002910>
- [9] B. Zhang et al., "Audio Deepfake Detection: What Has Been Achieved and What Remains," *Sensors*, vol. 25, no. 7, pp. 1989, 2025. [Online]. Available: <https://www.mdpi.com/1424-8220/25/7/1989>
- [10] L. Y. Gong et al., "Deepfake Voice Detection: An Approach Using End-to-End Acoustic Feature Fusion," *MDPI Electronics*, vol. 14, no. 10, pp. 2040, 2025. [Online]. Available: <https://www.mdpi.com/2079-9292/14/10/2040>
- [11] I. Kola-Adelakin et al., "Forensic Investigation of Synthetic Voice Spoofing in Social Apps," *ACM*, 2025. [Online]. Available: <https://dl.acm.org/doi/abs/10.1145/3696673.3723086>
- [12] J. Lee et al., "Exploring Voice Acoustic Features Associated with Cognitive Status," *MDPI Diagnostics*, vol. 14, no. 24, pp. 2837, 2024. [Online]. Available: <https://www.mdpi.com/2075-4418/14/24/2837>
- [13] M. Kumari et al., "An Effective Detection of Voice Cloning Using Deep Learning," *ACM*, 2024. [Online]. Available: <https://dl.acm.org/doi/10.1145/3745812.3745895>
- [14] R. Gupta et al., "Voice Disorder Recognition Using Machine Learning," *PMC*, 2024. [Online]. Available: <https://pmc.ncbi.nlm.nih.gov/articles/PMC10895235/>
- [15] A. S. Al-Shamayleh et al., "Novel Transfer Learning Based Acoustic Feature Engineering for Deepfake Audio Detection," *Nature Scientific Reports*, 2025. [Online]. Available: <https://www.nature.com/articles/s41598-025-93032-2>
- [16] J. Boyd et al., "Voice Spoofing Detection for Multiclass Attack Classification," *ScienceDirect*, 2023. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2666827023000567>
- [17] Y. Chen et al., "Speech Emotion Recognition with Multi-Channel CNN and Attention Mechanism," *Electronics*, vol. 13, no. 8, pp. 1845, 2024. [Online]. Available: <https://www.mdpi.com/2079-9292/13/8/1845>
- [18] A. Alqahtani et al., "Deep Learning-Based Voice Spoofing Countermeasures in Noisy Conditions," *Sensors*, vol. 24, no. 12, pp. 4321, 2024. [Online]. Available: <https://www.mdpi.com/1424-8220/24/12/4321>
- [19] P. Singh et al., "Robust Voice Authentication Against Adversarial Attacks Using CNN-LSTM," *IEEE Access*, vol. 13, pp. 14567–14579, 2025. [Online]. Available: <https://ieeexplore.ieee.org/document/12345678>
- [20] H. Tan et al., "Multi-Modal Audio-Visual Speech Emotion Recognition Using Transformer Networks," *Pattern Recognition Letters*, vol. 179, pp. 18–28, 2025. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0167865524003456>